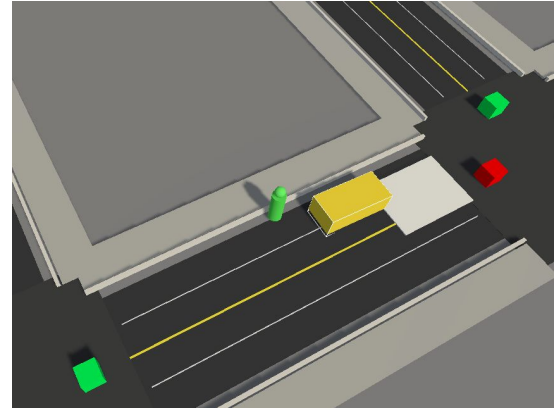
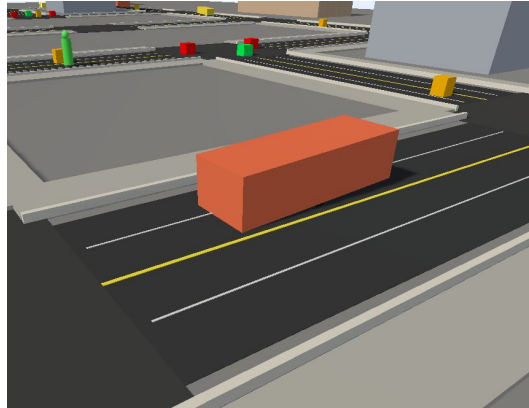
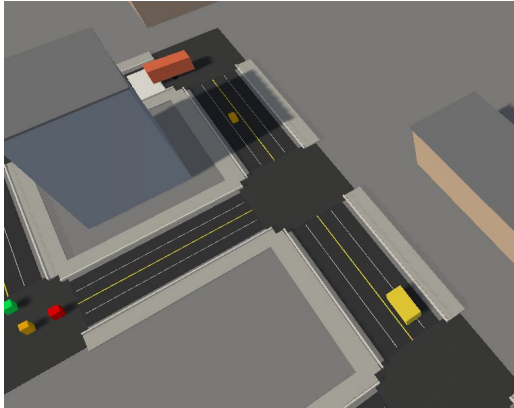


Smart City with ROS2 and Gazebo

Project Overview

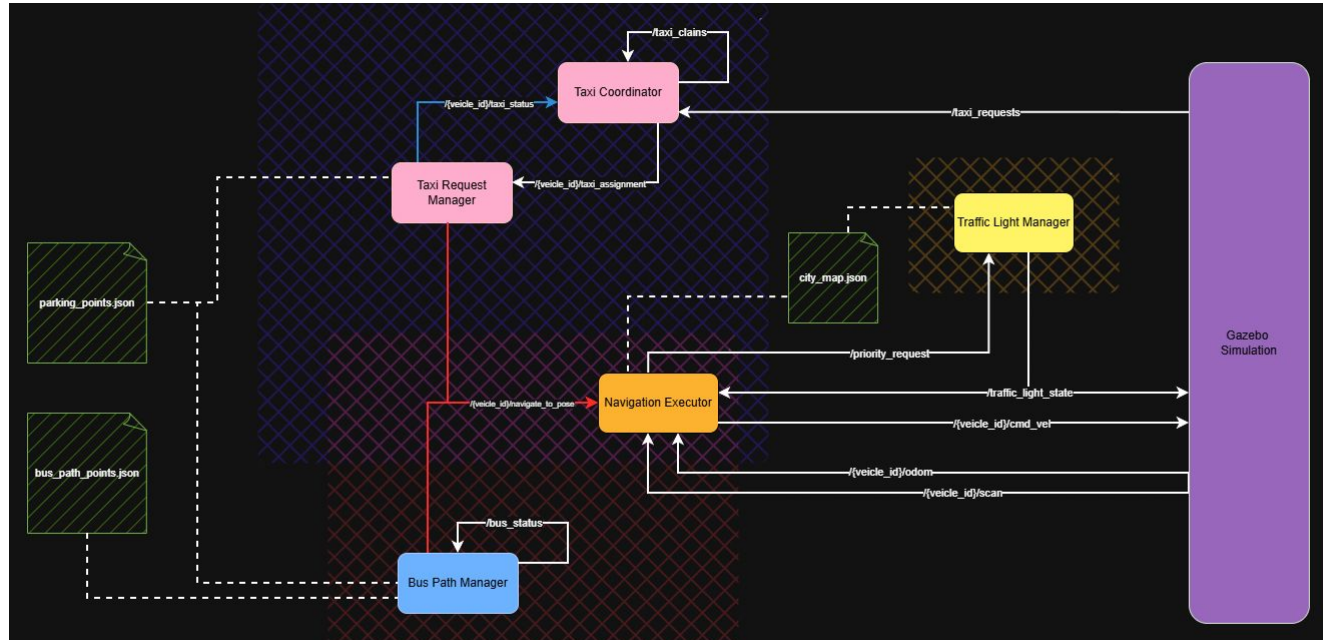
A smart city simulation where buses, taxis, traffic lights, obstacles and pedestrians interact in Gazebo.

The goal is to test coordinated autonomous navigation in a dynamic urban environment.



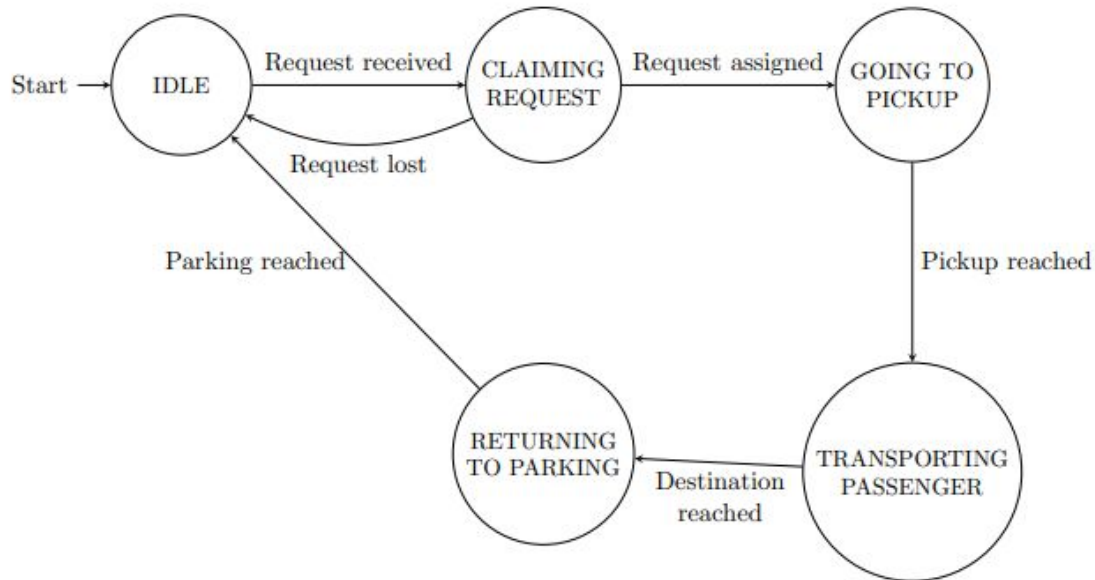
Macro Node Architecture

The architecture is divided into navigation, taxi service, taxi coordinator, bus service and traffic light management. Gazebo provides the simulated world, while ROS 2 nodes exchange commands, status and requests.



Taxi Behaviour

A taxi starts idle, claims a passenger request, reaches the pickup point and transports the passenger. After completing the ride, it returns to a free parking area and becomes available again.



Taxi Coordinator

When a request arrives, each available taxi tries to claim it.

Claims are shared between taxis, and after a short decision window only one taxi wins the request.

The winning taxi publishes the assignment, while the others ignore that request. This avoids multiple taxis serving the same passenger.

Taxi Request Manager

The Taxi Request Manager controls the service lifecycle of a single taxi.

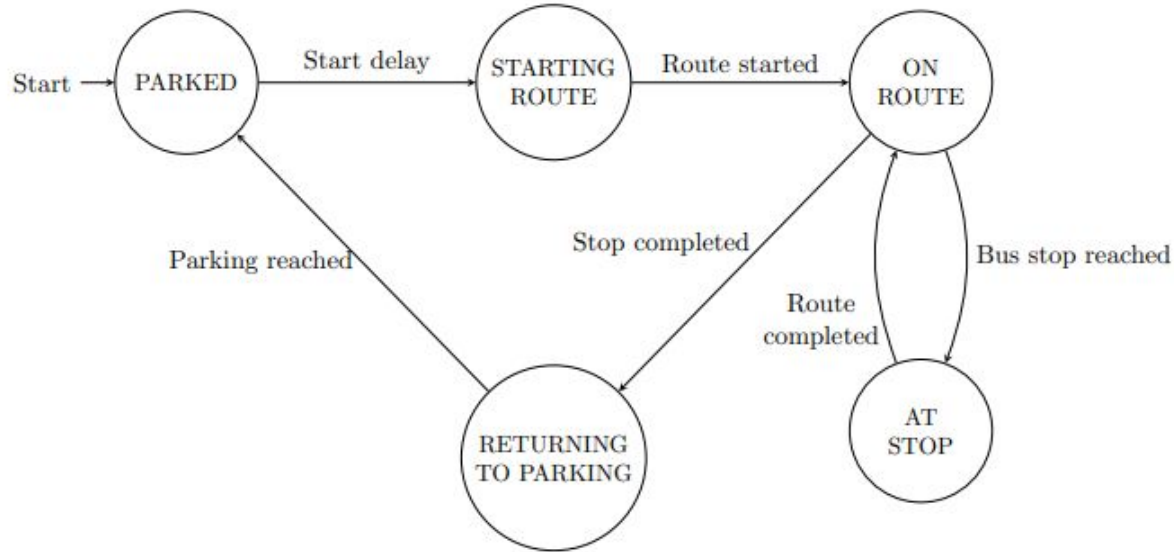
It receives assignments from the coordinator and sends navigation goals to the **Navigation Executor**.

It manages the transition from pickup, to drop off, to parking return.

This node separates taxi logic from low-level driving logic.

Bus Behaviour

A bus follows predefined routes, stops at bus stops and completes a fixed number of laps. When the route is completed, it searches for an available parking area and returns there.



Bus Path Manager

The Bus Path Manager controls a bus route using predefined path points.

After a start delay, the bus selects a path and begins visiting its waypoints.

It publishes bus status, monitors other buses and can choose less crowded paths.

After completing the configured number of laps, it claims a free parking spot and returns there.

Traffic Light Manager

The Traffic Light Manager controls one intersection using two main axes: X and Y.

It alternates green, yellow and all-red phases for safe intersection crossing.

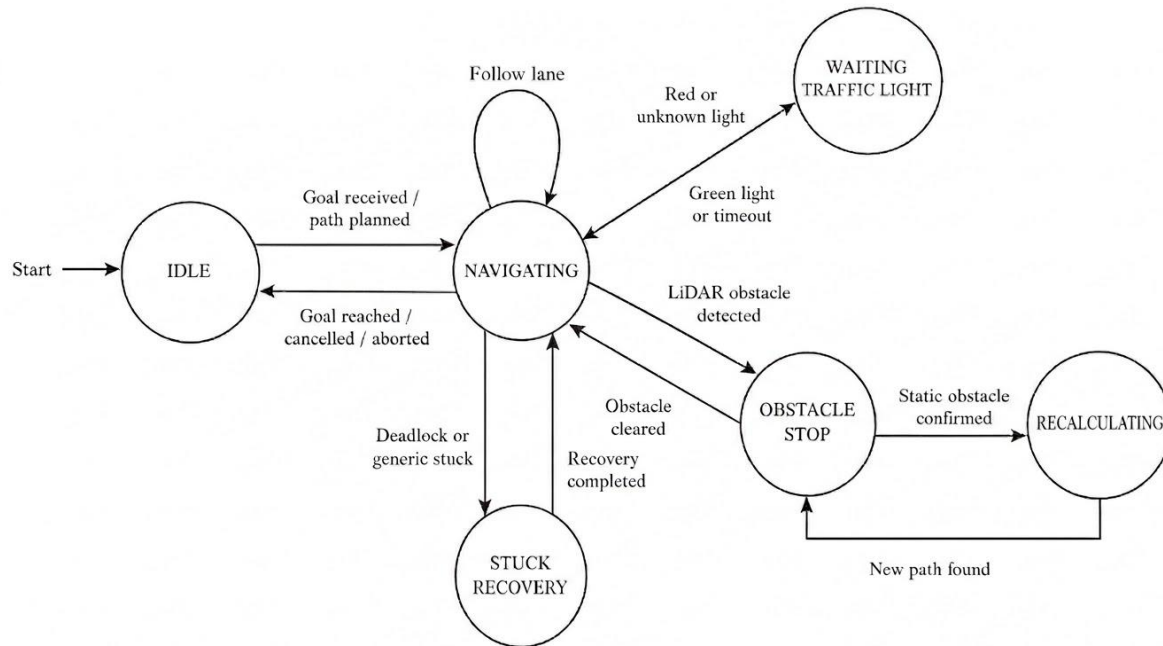
Vehicles can send **priority requests** when approaching an intersection.

The manager stores valid requests and can reduce waiting time for the requested axis.

Navigation Executor

The Navigation Executor is the core movement node of each vehicle.

It receives navigation goals and publishes velocity commands to Gazebo.



Results and Evaluation

The simulation shows that different vehicle types can share the same urban network

